

Grandperret Alexi

07 82 45 61 80 | alexi.grandperret@ipsa.fr | [Personal website](#)



[LinkedIn Profile](#)

EDUCATION

Institut Polytechnique des Sciences Avancées (IPSA)

Toulouse, FR

Fluid mechanics (turbomachinery, thermodynamics, turbulence, gas dynamics).

Sept. 2022 – Sept. 2027

Propulsion (turboprop, electric, nuclear, rocket). **Structures** (linear elasticity, finite elements).

LEAN manufacturing, quality management, aerospace law, accounting.

Universita Di Pisa

Pisa, IT

Aerospace dynamic systems, spacecraft technology, electric propulsion, rocket propulsion, space mechanisms and structures.

Aug. 2025 – Feb. 2026

EXPERIENCE

Assistant Director & Project Manager (GRUAU)

June 2026 – Sept. 2026

Gruau, Argentan site

Orne, FR

- Participated in **executive committee** meetings, assisted with site staff management, and tracked weekly site production and financial performance.
- Led a refrigerant fluid substitution project: thermal, thermodynamic and CFD analysis for a refrigerated .
- Set up 5S practices, continuously improved refrigerated compartment models and created manufacturing process sheets.

Production & Process Engineering Intern (Strate Composites)

June 2025 – Jul. 2024

Strate, Argentan site

Orne, FR

- Designed and implemented a vacuum gripping system and transport carts to improve material handling and production flow.
- Wrote manufacturing process sheets and technical documents to standardize composite parts production.
- Conducted on-site client visits to determine liability for defects and negotiate product deliveries.

Quality Manager (Magneti Marelli)

June 2023 – Jul. 2023

Marelli, Argentan site

Orne, FR

- Conducted problem-solving analyses (5D/8D) using Lean tools (5 Why, Ishikawa, Gemba Walk, Kaizen, 5S, Kanban, FIFO) to identify root causes and improve production flow.
- Supervised quality control, performed dimensional measurements on CMM and vacuum-based technical tests using oscilloscopes.
- Worked closely with production and engineering teams as well as clients to ensure quality and contribute to continuous process improvement.

PROJECTS

Rocket Project | Matlab, Simulink

2026 – present

- Digital twin of a rocket, thermodynamic and turbo-propulsion study, flight simulation, speed and trajectory parameterization.

Fred Et Rick Space | CATIA V5, <https://fredtrickspace.wixsite.com/aegis>

2025

- Designed a crewed spacecraft for 10 people in CATIA V5, including systems modeling and integration (visit the website for much more detail).

Drone Project | Matlab, Simulink

2025

- Implemented a closed-loop control system with PD controllers for 3D trajectory tracking.

SKILLS & INTERESTS

Languages: French (native), English (native), Spanish (fluent), Italian (B1), Portuguese (A2)

Software: Python, Matlab, Simulink, Nastran, Patran, Latex, CATIA V5, SolidWorks, STARCCM+

Activities: Music: piano (14 years), violin (3 years), orchestra and chamber music. Sports: tennis (6 years), soccer (12 years), kayaking (3 years), badminton (3 years) Travel: (9 countries), a year abroad in the United States as a freshman

Soft skills: open-mindedness, adaptability, teamwork, curiosity, communication, honesty, rigor.